

Internal Knowledge Assistant (IKA)

A cloud-based internal AI assistant designed to provide accurate, document-grounded answers to organizational knowledge while ensuring data security and scalability.

This project demonstrates systems thinking, requirements analysis, and cloud-native architecture design aligned with Technical Business Analyst, Solutions Engineer, and future AI Product Manager roles.

Problem Statement

Employees often spend excessive time searching across fragmented internal documentation (policies, onboarding materials, system docs), resulting in productivity loss and inconsistent information retrieval.

Solution Overview

The Internal Knowledge Assistant (IKA) is a cloud-native AI system that:

- Retrieves relevant internal documents using semantic search
- Generates grounded responses using retrieval-augmented generation (RAG)
- Provides citations to ensure transparency and trust
- Is designed with scalability, security, and reliability in mind

Key Features (MVP)

- Natural-language question answering
- Document ingestion and embedding
- Vector-based semantic retrieval
- LLM-generated answers with citations
- Basic logging and monitoring

High-Level Architecture

![[System Architecture Diagram]](diagrams/SysArchitecture.drawio.png)

Tech Stack (Initial)

- Backend: Python, FastAPI
- Embeddings: OpenAI Embeddings
- Vector Database: FAISS (local, MVP)
- LLM: OpenAI API

- Storage: Local filesystem (S3 planned)
- Cloud Target: AWS (future deployment)

Learning & Career Objectives

This project is designed to demonstrate:

- Translation of business requirements into technical systems
- Cloud and AI system design trade-offs
- Understanding of modern DevOps, scalability, and reliability concepts
- Clear technical communication for stakeholders

Project Status

- ☒ Requirements definition
- ☒ Architecture design
- ☐ Backend API implementation
- ☐ Document ingestion pipeline
- ☐ Retrieval-augmented QA
- ☐ Cloud deployment

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